PulmoAI: A Multi-Agent Clinical Decision Support System for Chest CT Analysis

Team: Med-Squad

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ABSTRACT

Problem	Lung cancer is the #1 cause of cancer death — yet diagnosis relies on scarce specialist expertise
Gap	Current AI diagnostic tools address only a single specialty, offering no integrated clinical view from a single scan.
Solution	PulmoAI: a multi-agent AI system that analyzes chest CT scans across 5 specialties in one pipeline
Method	MedGemma + LoRA · VoxTell auto-segmentation · LangGraph conversational agent
Deployment	Full-stack web app — accessible without specialist hardware

RELATED WORK

System	Specialty	System Capabilities
CheXNet	Radiology only	Radiology Text Only (No segmentation/dialogue)
Med-PaLM 2	General medical Q&A	Text Q&A Only (Not grounded in patient's scan)
LLaVA-Med	Medical VQA	Single-turn VQA (No clinical workflow)
nnU-Net / SAM-Med	Segmentation only	Pixel Segmentation Only (No report generation)
PulmoAI (Our)	**Radiology + Cardiology + Oncology + Findings + Impression + Segmentation**	**Full End-to-End Pipeline**

OBJECTIVES

- **Build a 5-agent AI pipeline for multi-specialty chest CT analysis**
- **Automate CT segmentation guided by the generated report**
- **Enable clinician-facing dialogue grounded in each patient's own results**
- **Deploy as a production web application — no specialist hardware required**

BUSINESS CONTEXT & DEPLOYMENT MODEL

Model	B2B — architecture-as-a-service for healthcare providers
How it works	Hospital provides their own clinical CT data → PulmoAI adapts the pipeline via fine-tuning
Why adaptable	Base model (MedGemma) is general-purpose — must be fine-tuned on hospital-specific data to reflect local patient population & terminology
This project	OpenM3Chest, CT-RATE used as simulated hospital dataset — MedGemma fine-tuned from scratch on it to demonstrate the adaptation pipeline

DATASET & PRE-PROCESSING & FINE-TUNING

DATASET OVERVIEW: OPENM3CHEST

Source	OpenM3Chest — derived from NLST (National Lung Screening Trial)
CT scan	Original DICOM (.dcm) → preprocessed to NumPy arrays in dataset
Labels & clinical data	JSON — demographics, smoking history, VQA questions, answers
Total tasks	**15 labeled tasks** across **3 clinical domains**
Scope	**Radiology · Cardiology · Oncology — no labels for Interpretation or Verification**

OPENM3CHEST: TASK BREAKDOWN

Agent	Task	Output type
Radiology	Chest abnormality (×7)	Classification label
	Nodule presence	Yes / No
	Nodule attenuation	Solid / Ground-glass / Mixed
	Nodule location	Lobe (RUL, RLL, LUL...)
	Nodule margin	Spiculated / Smooth / Lobulated
	Nodule size	Measurement category
Cardiology	CVD diagnosis	Classification label
	CVD mortality risk	Low / High
Oncology	Lung cancer risk	No cancer / Cancer within follow-up

Radiology — Screening

Atelectasis	Is there any atelectasis, segmental or greater?	No
Pleural Effusion	Is there any pleural thickening or effusion?	Yes
Hilar / Mediastinal Mass	Is there a non-calcified mass or adenopathy ≥10mm in the hilar/mediastinal region?	No
Chest Wall Abnormality	Is there any chest wall abnormality (bone destruction or metastasis)?	No
Consolidation	Is there any consolidation?	No
Emphysema	Is there any emphysema?	No
Fibrosis / Honeycombing	Is there any fibrosis, honeycombing, reticular/reticulonodular opacities, or scarring?	No
Nodule Present	Is there any lung nodule?	Yes

Radiology — Nodule Detail

Size — 4~6mm
What is the nodule size?

Margin — Smooth
What is the nodule margin type?

Location — Right Lower Lobe
In which lobe is the nodule located?

Attenuation — Solid
What is the predominant attenuation of the nodule?

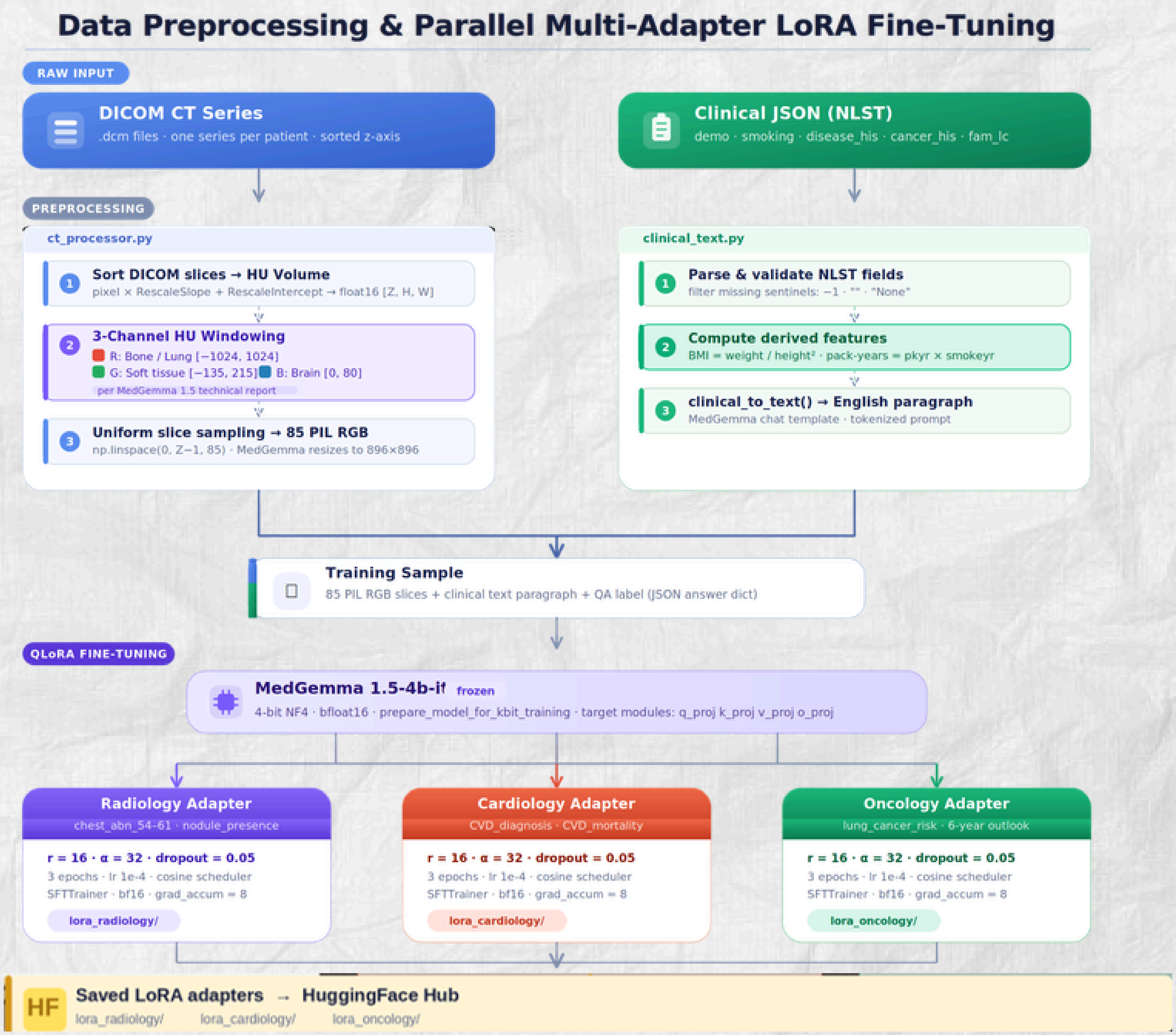
Cardiology

CVD Diagnosis — Yes
Can you identify any notable abnormalities in the cardiovascular system?

CVD Mortality Risk — Low risk
Predict the risk of cardiovascular disease mortality.

Oncology

Lung Cancer Risk — No cancer within follow-up
Given the current indicators, what's the six-year outlook for developing lung cancer?


Data Preprocessing & Parallel Multi-Adapter LoRA Fine-Tuning
RAW INPUT
DICOM CT Series
.dcm files · one series per patient · sorted z-axis
Clinical JSON (NLST)
demo · smoking · disease_his · cancer_his · fam_lc
PREPROCESSING
ct_processor.py
1 Sort DICOM slices → HU Volume
pixel × RescaleSlope + RescaleIntercept → float16 [Z, H, W]
2 3-Channel HU Windowing
R: Bone / Lung [−1024, 1024]
G: Soft tissue [−135, 215]
B: Brain [0, 80]
per MedGemma 1.5 technical report
3 Uniform slice sampling → 85 PIL RGB
np.linspace(0, Z−1, 85) · MedGemma resizes to 896×896
clinical_text.py
1 Parse & validate NLST fields
filter missing sentinels: −1 · "" · "None"
2 Compute derived features
BMI = weight / height² · pack-years = pkyr × smokeyr
3 clinical_to_text() → English paragraph
MedGemma chat template · tokenized prompt
Training Sample
85 PIL RGB slices + clinical text paragraph + QA label (JSON answer dict)
QLoRA FINE-TUNING
MedGemma 1.5-4b-it frozen
4-bit NF4 · bfloat16 · prepare_model_for_kbit_training · target modules: q_proj k_proj v_proj o_proj
Radiology Adapter
chest_abn_54-61 · nodule_presence
r = 16 · α = 32 · dropout = 0.05
3 epochs · lr 1e-4 · cosine scheduler
SFTTrainer · bf16 · grad_accum = 8
lora_radiology/
Cardiology Adapter
CVD_diagnosis · CVD_mortality
r = 16 · α = 32 · dropout = 0.05
3 epochs · lr 1e-4 · cosine scheduler
SFTTrainer · bf16 · grad_accum = 8
lora_cardiology/
Oncology Adapter
lung_cancer_risk · 6-year outlook
r = 16 · α = 32 · dropout = 0.05
3 epochs · lr 1e-4 · cosine scheduler
SFTTrainer · bf16 · grad_accum = 8
lora_oncology/
HF
Saved LoRA adapters → HuggingFace Hub
lora_radiology/
lora_cardiology/
lora_oncology/

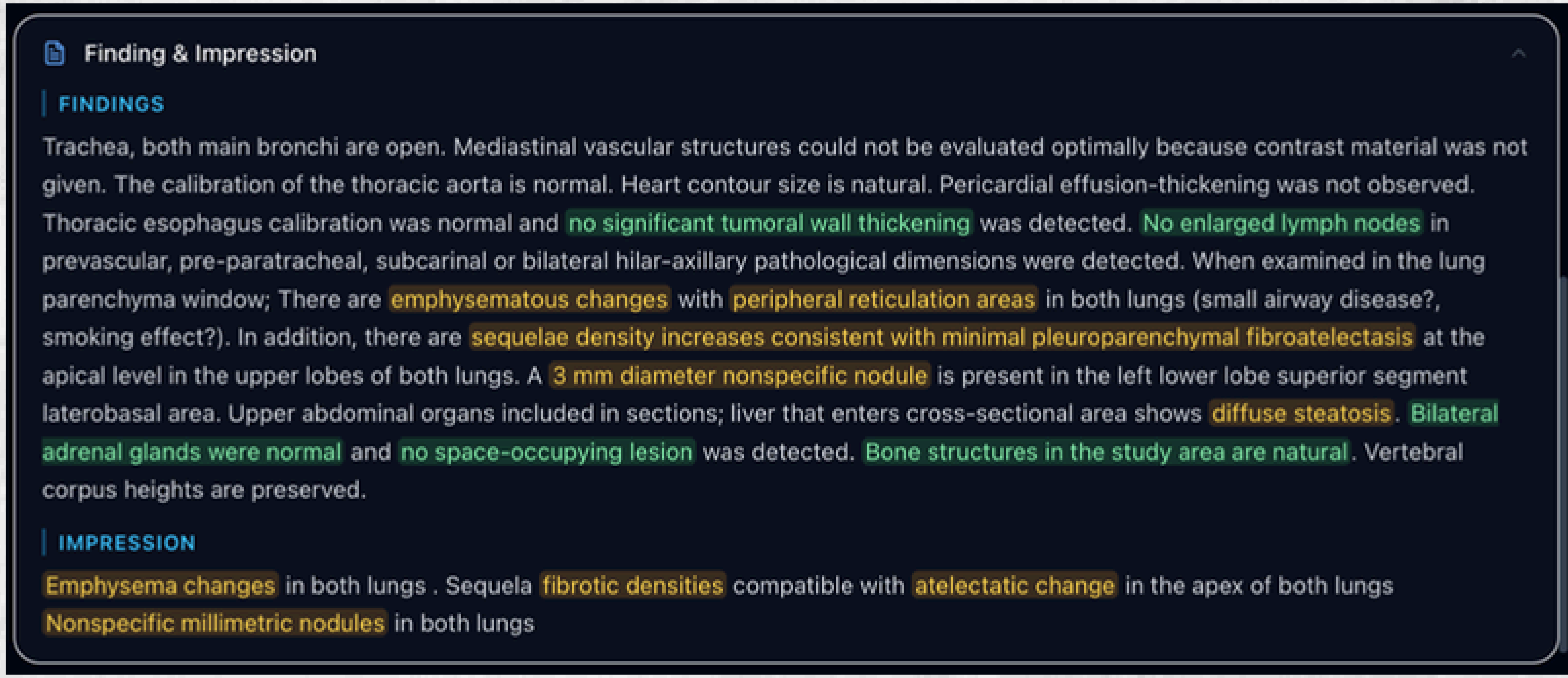
DATASET OVERVIEW: CT-RATE

Source	CT-RATE: Open source dataset for 3D CT-SCAN
CT scan	Original DICOM (.dcm) → preprocessed to NumPy arrays in dataset
Total tasks	**2 Tasks: Findings and Impressions**

NOTES:

- **Findings:** What the radiologist **sees** (observations, measurements, abnormalities on the scan).
- **Impressions:** What those findings **mean** (summary, final diagnosis, or next steps).

DATASET OVERVIEW: CT-RATE


Finding & Impression

FINDINGS

Trachea, both main bronchi are open. Mediastinal vascular structures could not be evaluated optimally because contrast material was not given. The calibration of the thoracic aorta is normal. Heart contour size is natural. Pericardial effusion-thickening was not observed. Thoracic esophagus calibration was normal and no significant tumoral wall thickening was detected. No enlarged lymph nodes in prevascular, pre-paratracheal, subcarinal or bilateral hilar-axillary pathological dimensions were detected. When examined in the lung parenchyma window; There are emphysematous changes with peripheral reticulation areas in both lungs (small airway disease?, smoking effect?). In addition, there are sequelae density increases consistent with minimal pleuroparenchymal fibroatelectasis at the apical level in the upper lobes of both lungs. A 3 mm diameter nonspecific nodule is present in the left lower lobe superior segment laterobasal area. Upper abdominal organs included in sections; liver that enters cross-sectional area shows diffuse steatosis. Bilateral adrenal glands were normal and no space-occupying lesion was detected. Bone structures in the study area are natural. Vertebral corpus heights are preserved.

IMPRESSION

Emphysema changes in both lungs . Sequela fibrotic densities compatible with atelectatic change in the apex of both lungs
Nonspecific millimetric nodules in both lungs

Why **CT-RATE** over **OPENM3CHEST?**

- **Limitation of OPENM3CHEST:** Restricted to simple binary classification (Yes/No labels).
- **The CT-RATE Advantage:** Provides rich, narrative data structured into clinical Findings and Impressions.
- **Clinical Impact:** Moves the AI from a basic classifier to a true diagnostic assistant, generating detailed reports that actively support radiologists in complex decision-making.

DATASET SPLIT FOR AGENT FINE-TUNING

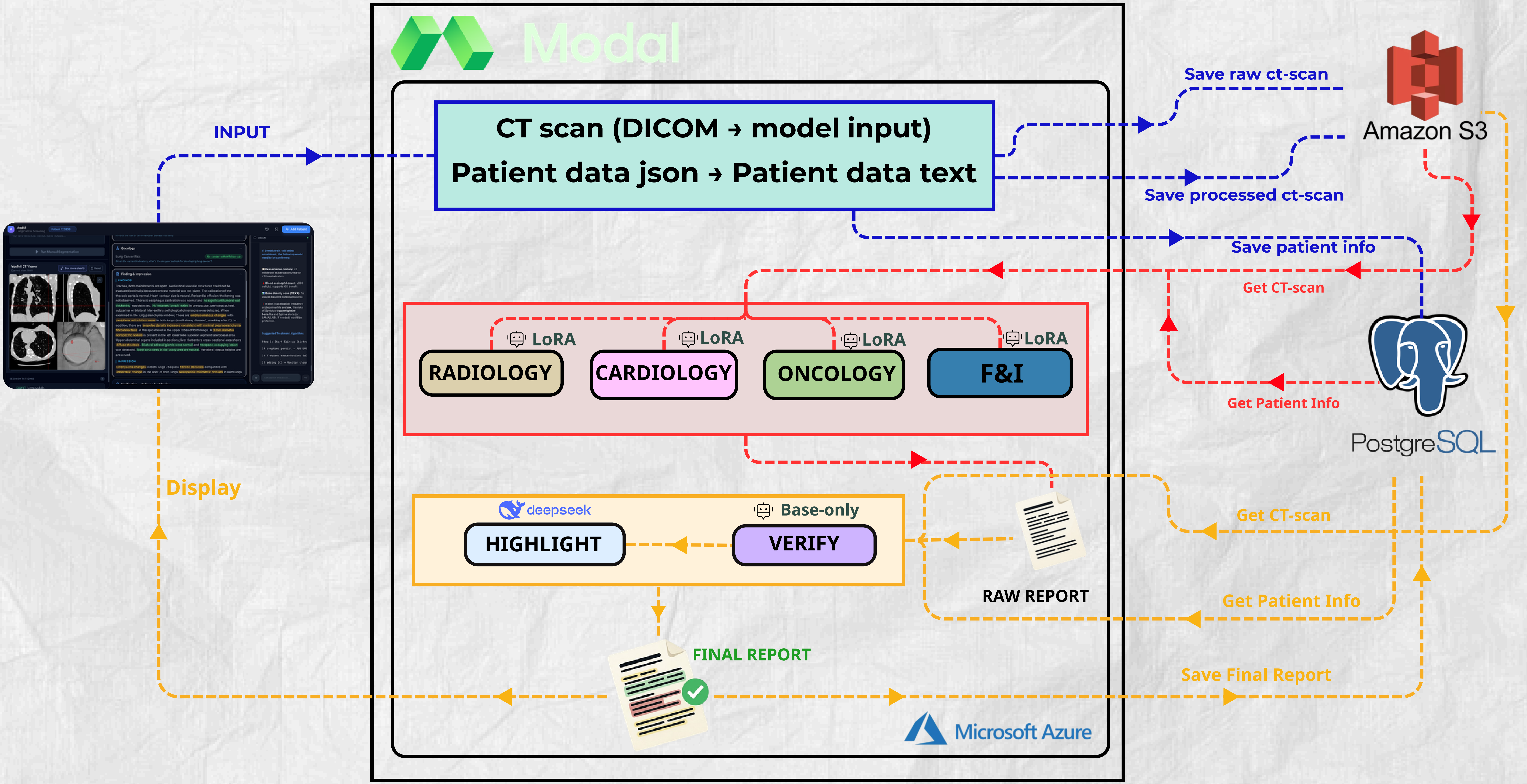
	OPENM3CHEST	CT-RATE
Train	Oncology: 500 Radiology: 500 Cardiology: 500	Findings and Impressions: 500
Test	Oncology: 200 Radiology: 200 Cardiology: 200	Findings and Impressions: 200

⚠️ Small subset used due to GPU resource constraints (A100 80GB required)

METHOD

Multi-Agent Clinical Report Generation Pipeline

MULTI-AGENT CLINICAL REPORT GENERATION PIPELINE



VERIFICATION AGENT

What it does

- Independently re-reads the raw report from all 4 agents
- Can disagree with specific findings
- Surfaces unreported findings missed by specialist agents
- Adds clinical caveats and differential considerations

How it works

- Uses base MedGemma — no LoRA adapter
- Intentionally unbiased: not fine-tuned toward any specialty

Output

- Verified report → passed to Highlight agent (DeepSeek) for color annotation

Verification — Independent Review

FINDINGS AND IMPRESSION

Chest

I agree with all points regarding the absence of atelectasis, pleural thickening/effusion, consolidations, parenchymal lesions like masses or nodules >1cm, bone destruction, etc., as well as the lack of obvious signs suggestive of severe pulmonary fibrosis such as extensive honeycombing or widespread reticular patterns beyond mild background interstitial markings potentially related to chronic inflammation from her known respiratory conditions. Emphysematous changes appear prominent bilaterally involving primarily the apices but also extending into more basal regions. Some patchy groundglass opacity might represent superimposed inflammatory processes rather than definite edema. Small calcifications along bronchovascular bundles may reflect benign granulomas. **I disagree with the finding "Is there any consolidation?" → Yes.** While some subtle hazy increased density exists particularly towards the right middle lobe base possibly representing early infectious process versus postinflammatory changes, definitive lobar consolidation isn't clearly visualized across multiple phases if available. However, small foci cannot entirely rule out very early pneumonia especially considering patient symptoms. If clinical suspicion remains high despite negative imaging, further evaluation including lateral decubitus views or short interval repeat exam would help clarify. Also, consider correlation with laboratory data.

Cardiovascular

I agree with the assessment showing unremarkable cardiac silhouette without pericardial fluid accumulation nor aortic dissection features. Normal caliber vessels throughout.

Oncology Risk

Based solely on these static images presented here, while she has several factors increasing potential future malignancy risk due to heavy smoking history and possible underlying COPD contributing to impaired clearance mechanisms, currently nothing specific suggests imminent development of primary lung cancer compared to general population risks unless additional information becomes available suggesting otherwise. This interpretation should always account for limitations inherent in assessing long term prognosis purely through single snapshot scans.

Imaging Report Accuracy

I largely agree with the radiology summary section describing the overall picture including emphysema pattern, bibasilar linear scar tissue likely reflecting old pneumonic infiltrates leading to atelectasis ("sequelae density increase"), and the solitary tiny (<6mm) solid nodule noted in the left lower lobe periphery. It accurately reflects common incidental findings often seen in patients with advanced obstructive airways diseases who smoke heavily.

Unreported Findings

Besides the aforementioned minor considerations about possibility of resolving infection vs postinflammation causing slight haziness near RML base, another observation includes scattered punctate micronodules distributed diffusely throughout both lungs measuring up to ~3mm maximum dimension. These can sometimes be associated with various etiologies ranging from simple intrapulmonary hemorrhage secondary to vasculitis affecting smaller vessels, metastatic seeding although less likely given their appearance, or even reactive lymphoid hyperplasia. Their significance depends greatly upon context and prior medical history. Another noteworthy feature is marked tortuousity and atherosclerotic plaque burden evident in descending thoracic aorta and proximal abdominal segments shown partially imaged. Mild degenerative discogenic spurring is apparent lumbar spine levels.

FUNCTION: CHATBOT

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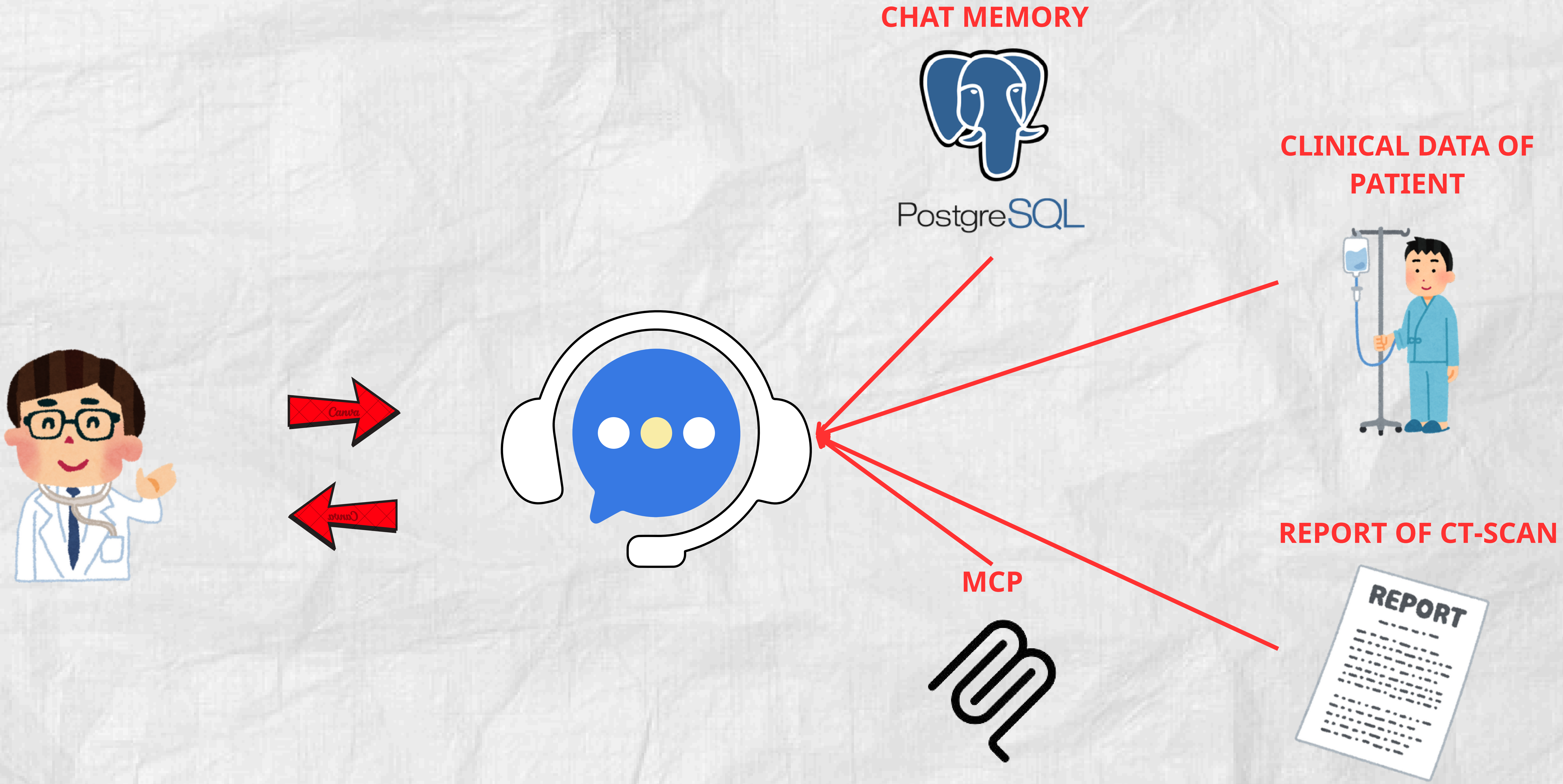
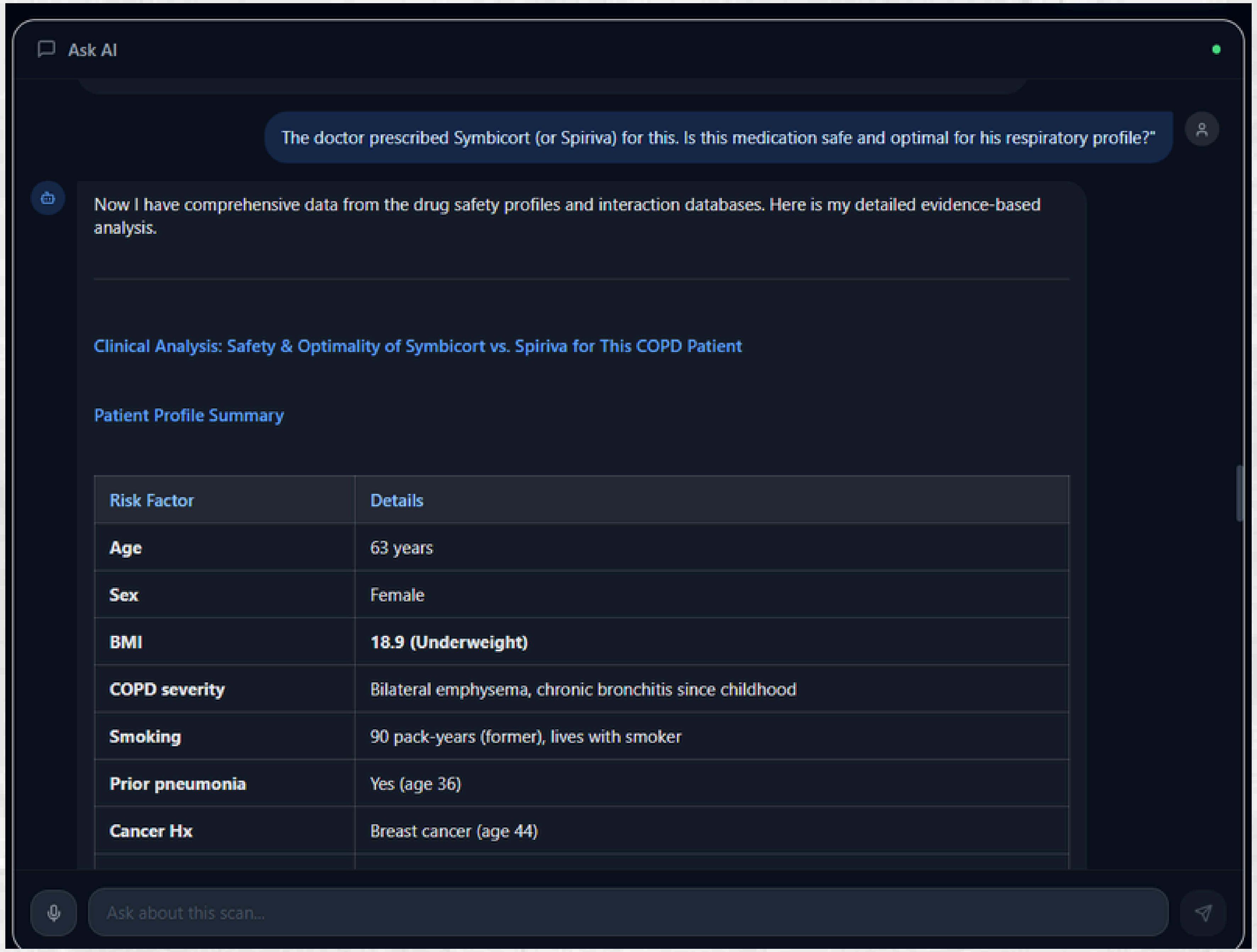


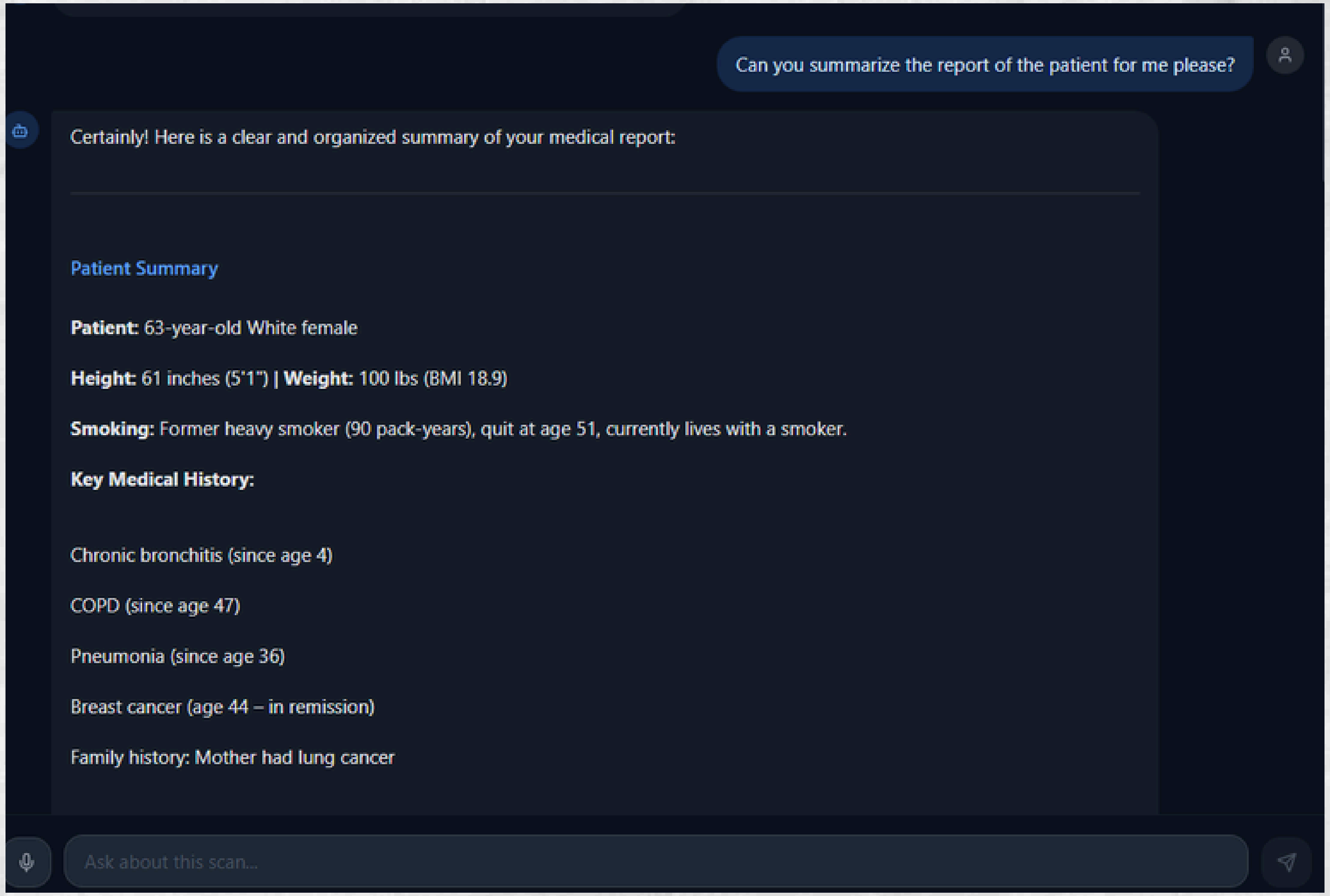
- **Drug Interactions Tool:** Identifies risk factors when mixing medications.
- **Warnings & Cautions Tool:** Extracts safety alerts, contraindications, and clinical precautions.

FUNCTION: CHATBOT

Drug Interaction



Summarization



FUNCTION: SEGMENTATION

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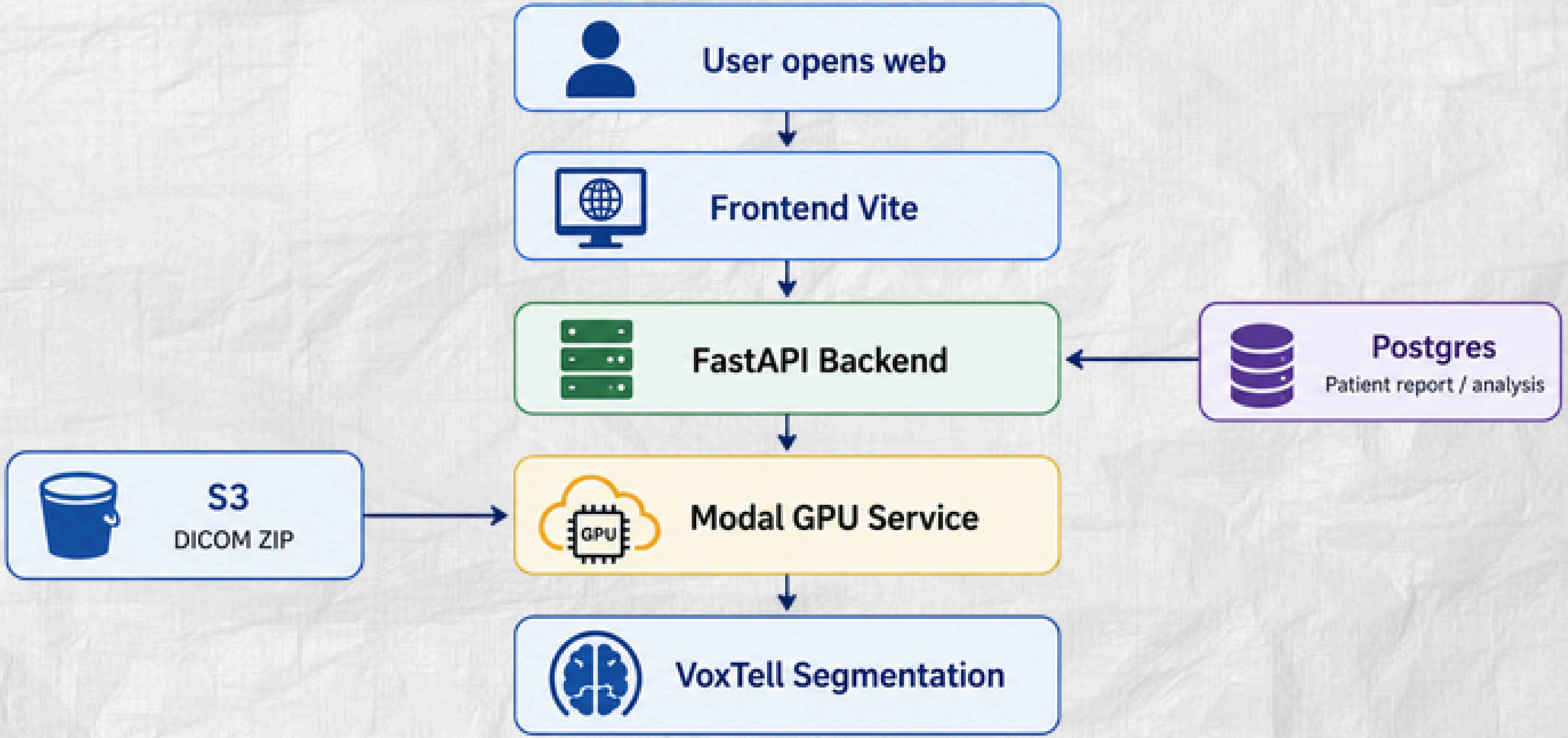
System Overview:

- The VoxTell Web system allows doctors to view CT scans on a web interface and automatically generate an initial segmentation mask based on the patient report.

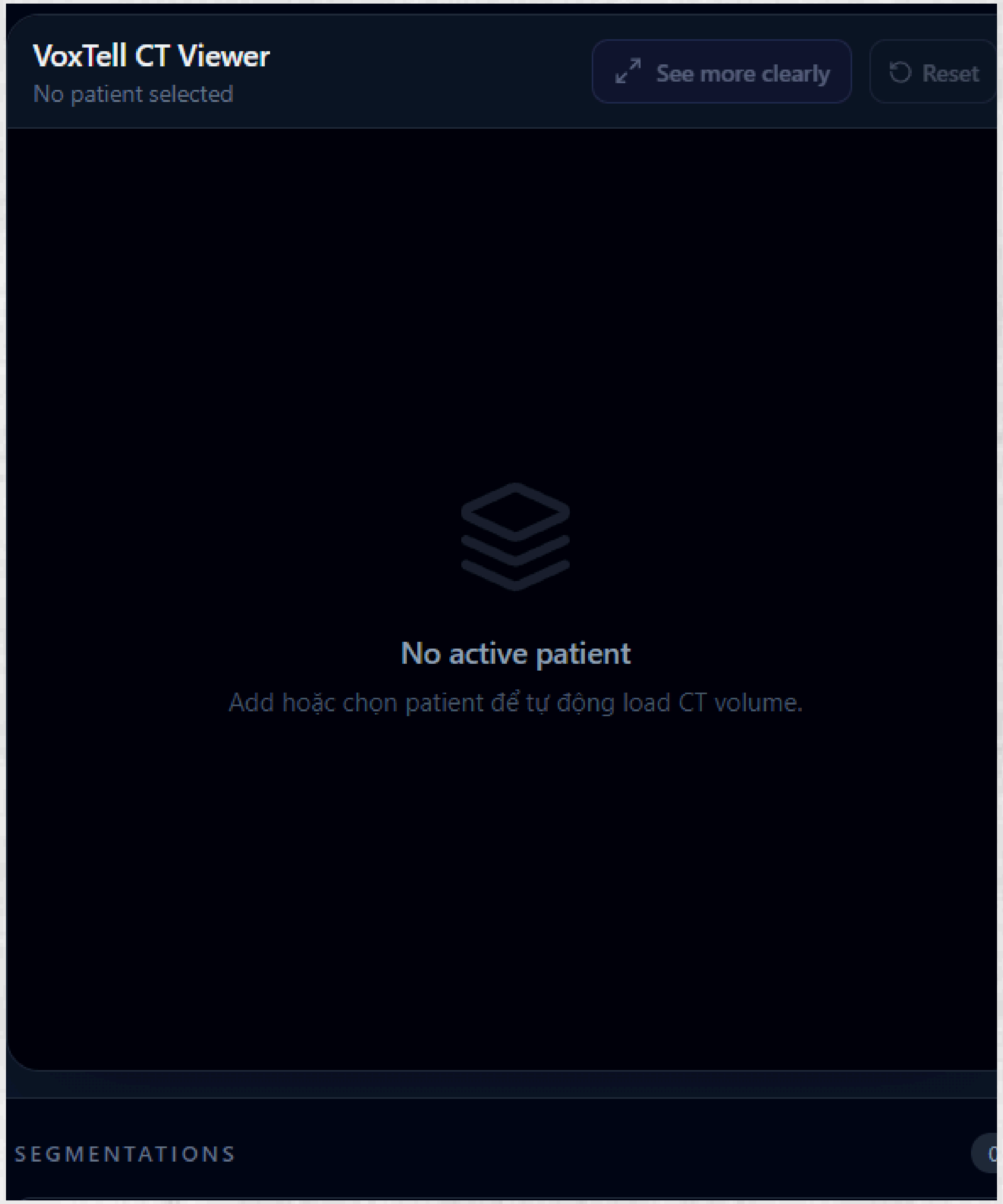
Component	Role
Frontend Vite	Displays CT volume and segmentation masks
FastAPI Backend	Handles API requests and coordinates the workflow
Modal GPU	Runs VoxTell segmentation with GPU acceleration
S3 + Postgres	Stores DICOM data and patient analysis reports

Overall Architecture

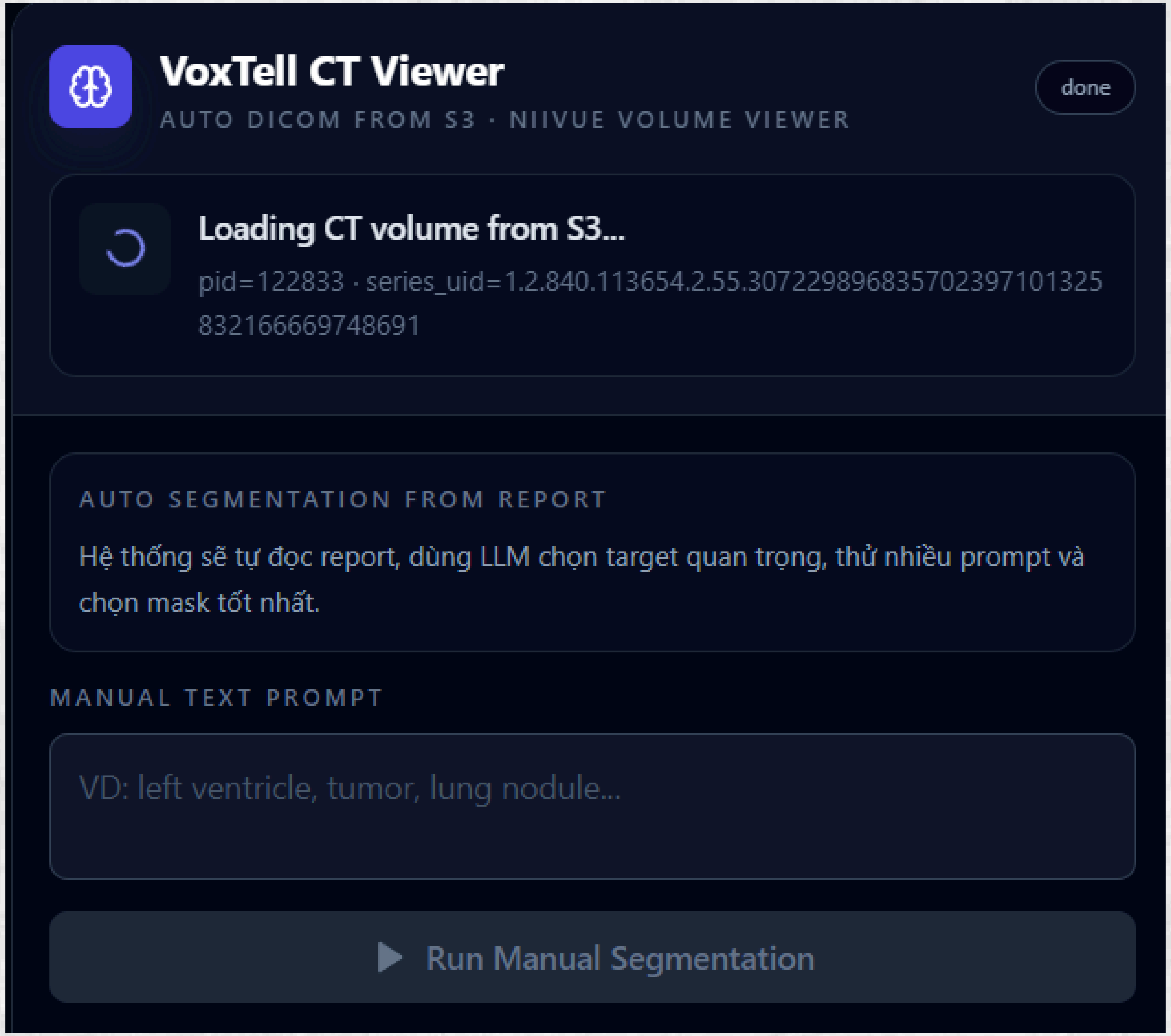


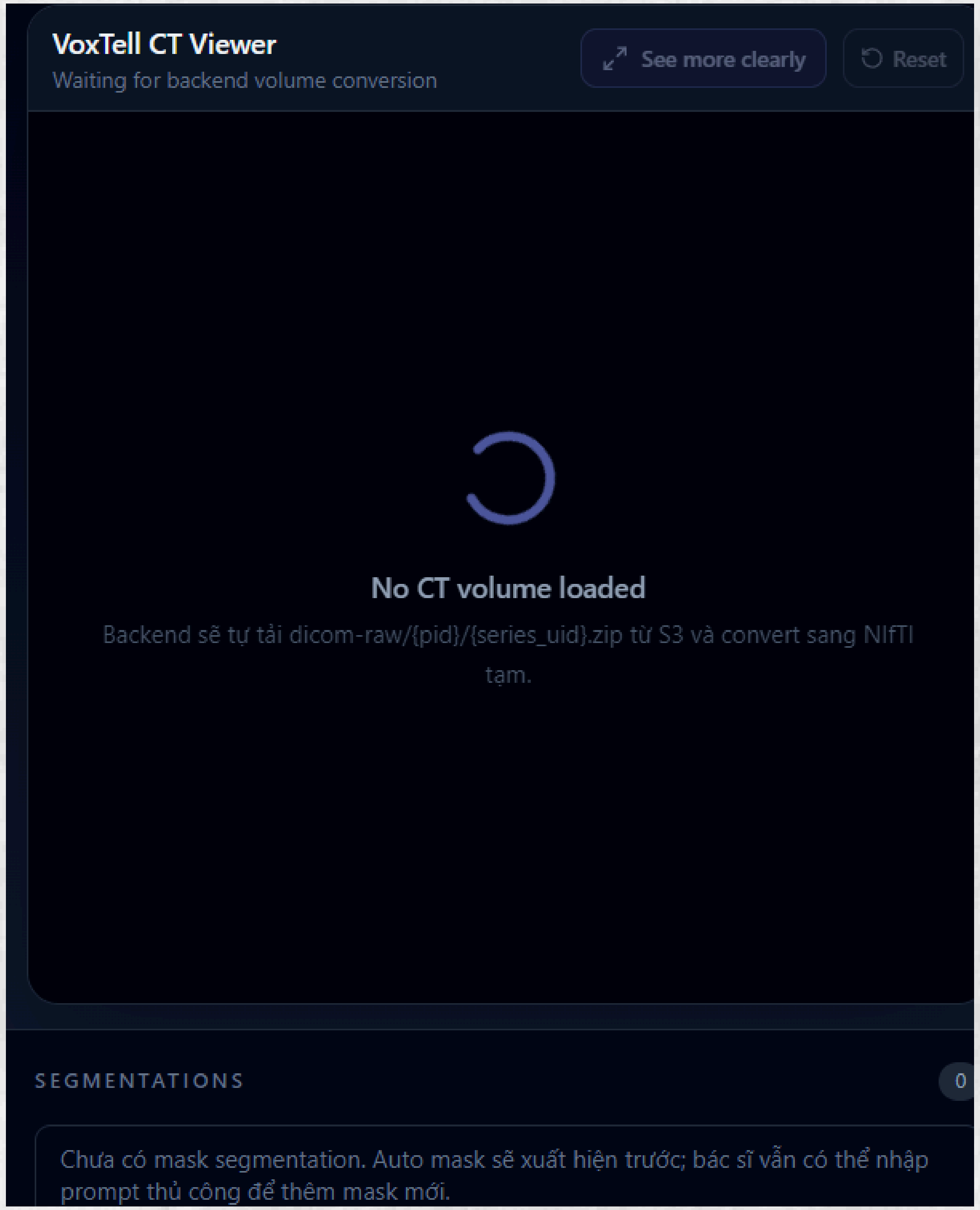
Local deployment: Backend proxies requests to Modal via VOXTELL_MODAL_BASE_URL

FUNCTION: SEGMENTATION



FUNCTION: SEGMENTATION





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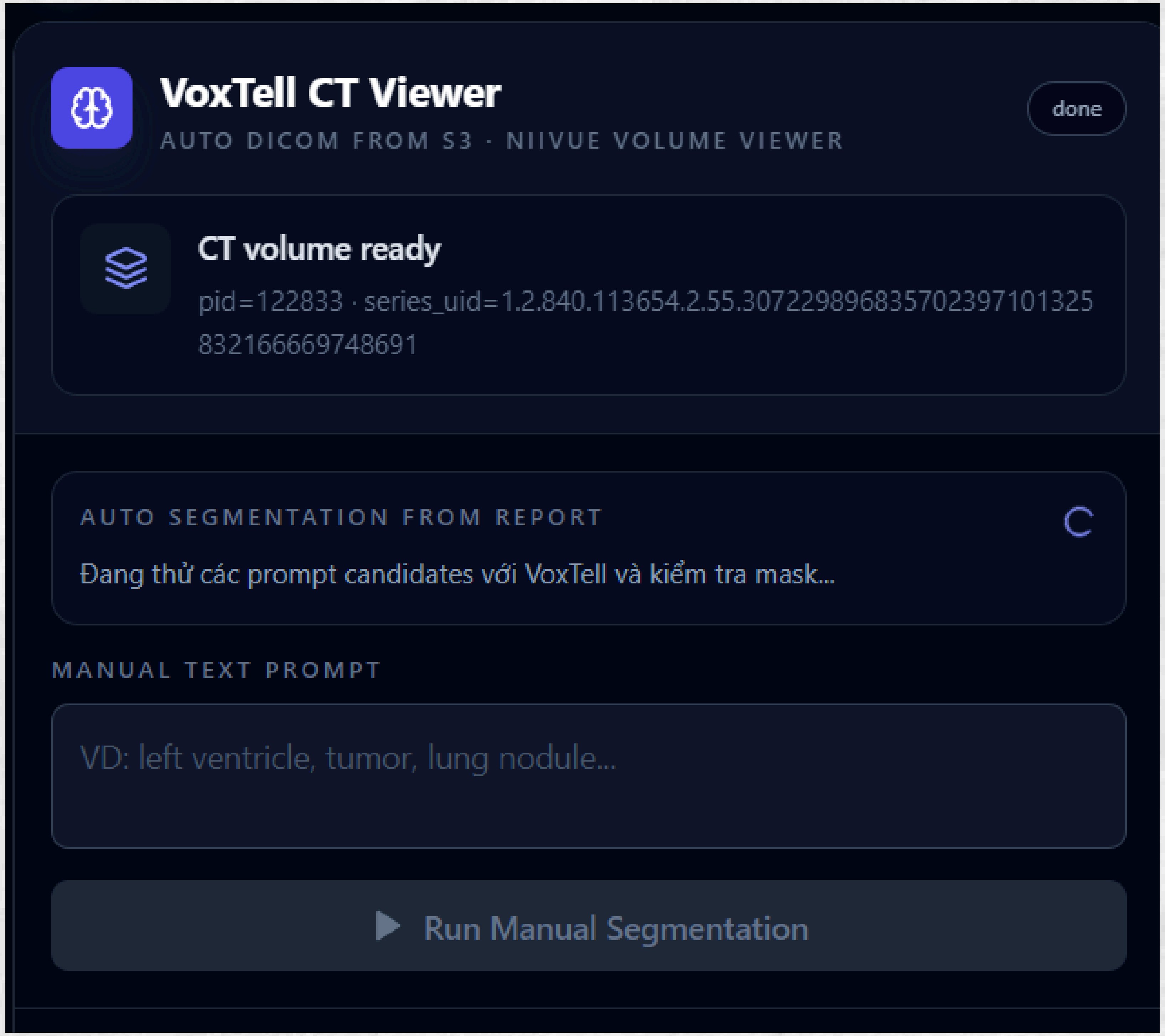
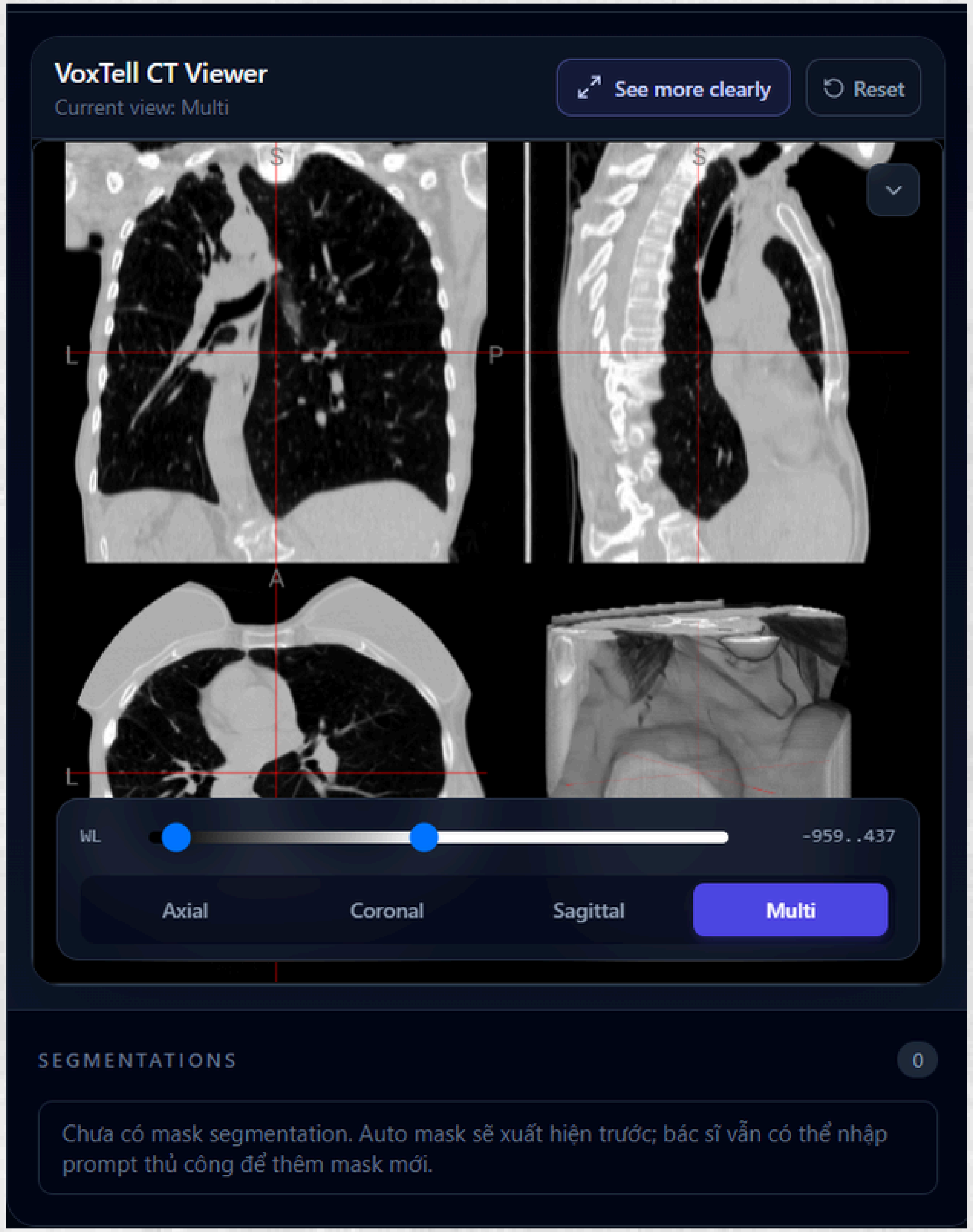
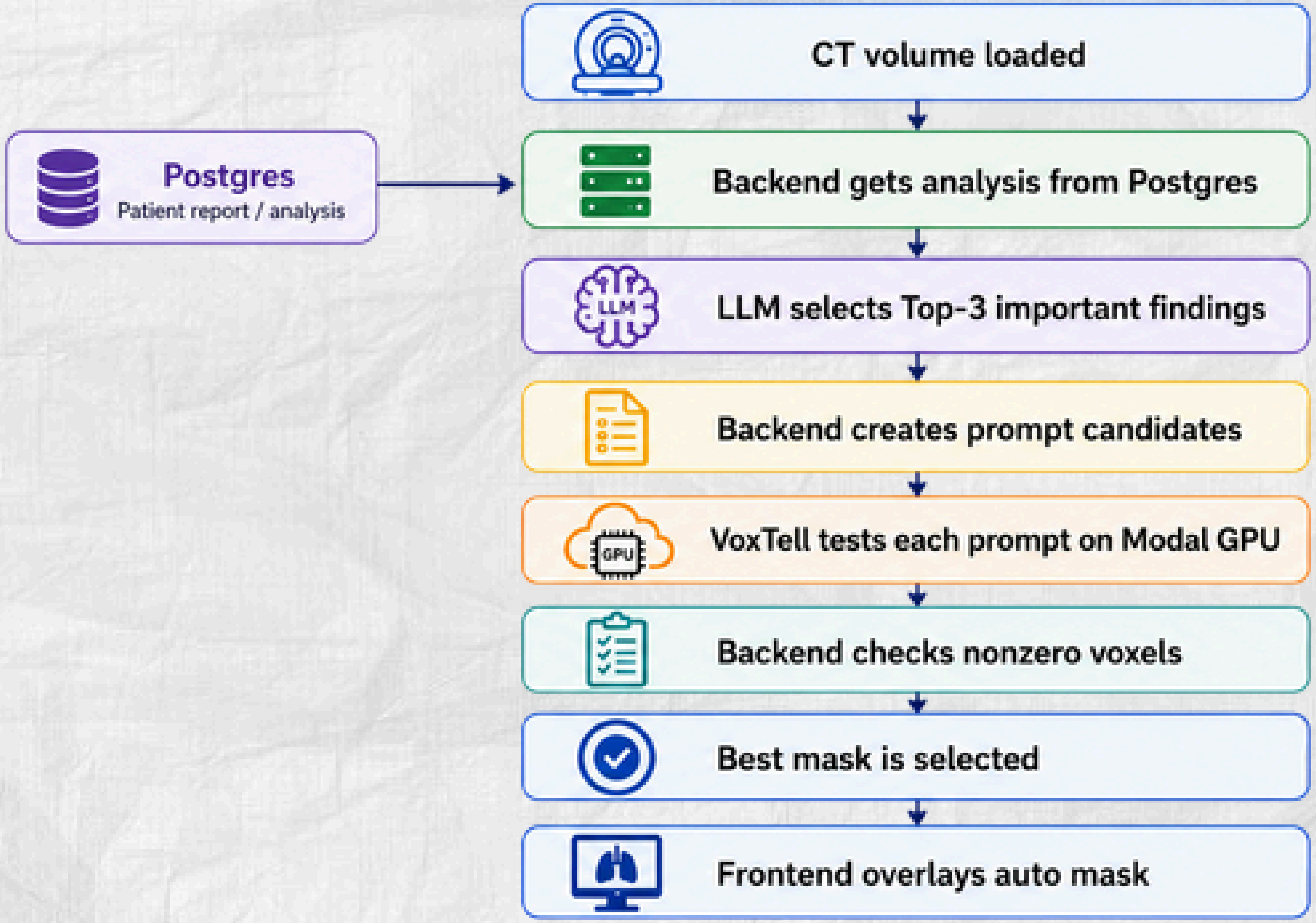




Auto Segmentation Workflow



Example

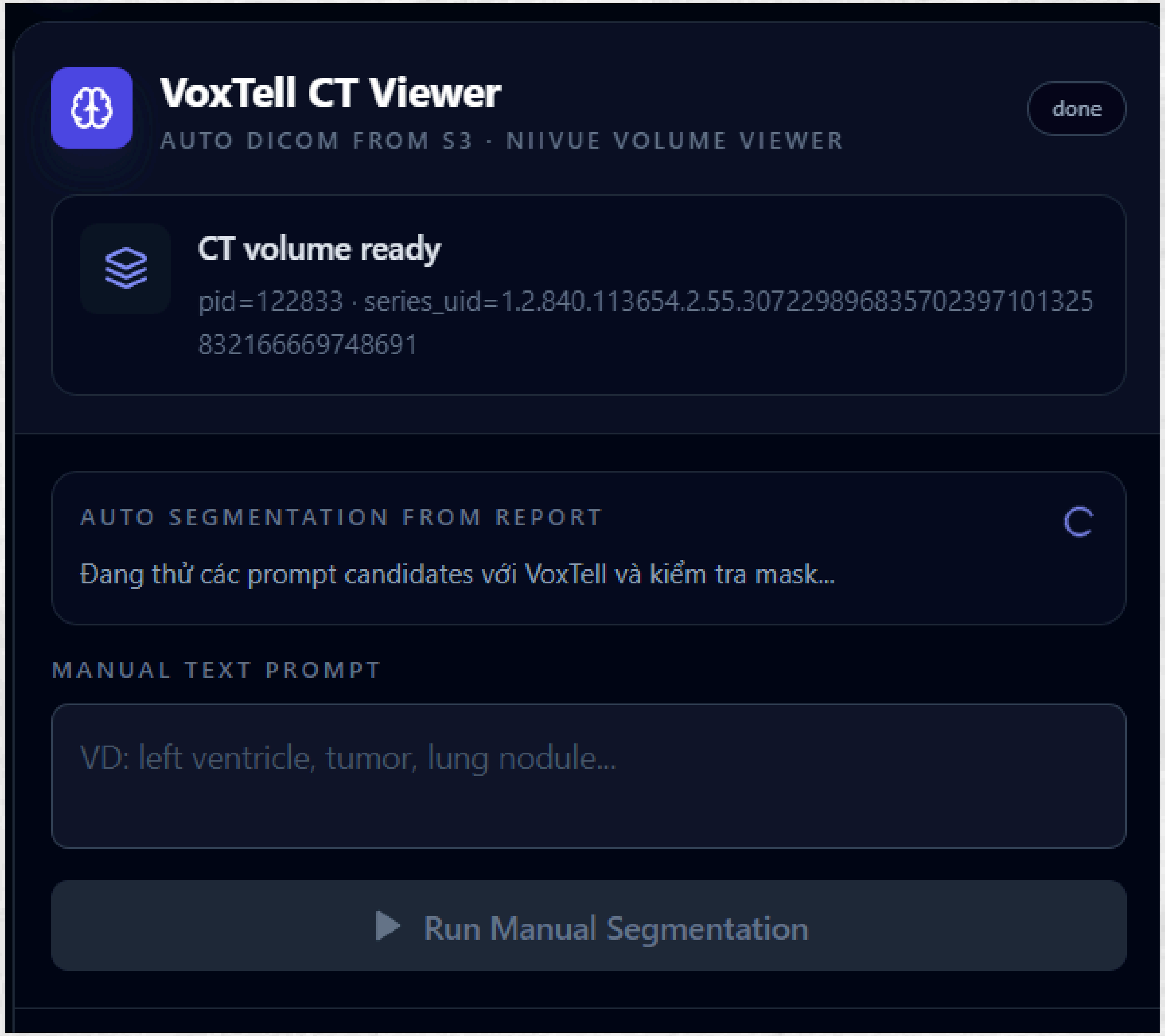
- Top-3 findings
  - lung nodule
  - pulmonary nodule
  - right lower lobe lung nodule

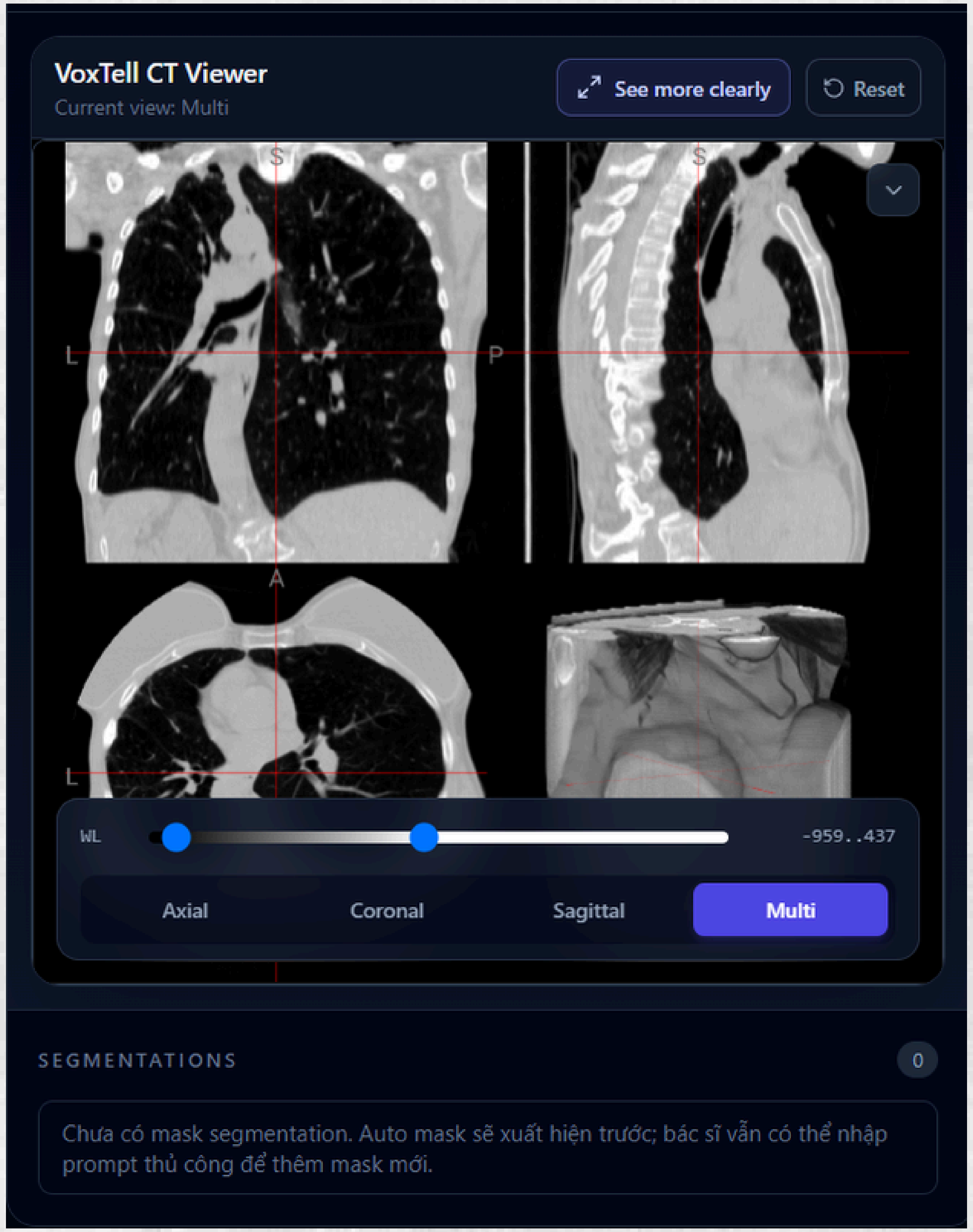
Selected prompt: lung nodule

Validation: nonzero voxels > 0

Auto segmentation provides an initial suggestion. Doctors can still run **manual prompts.**

FUNCTION: SEGMENTATION





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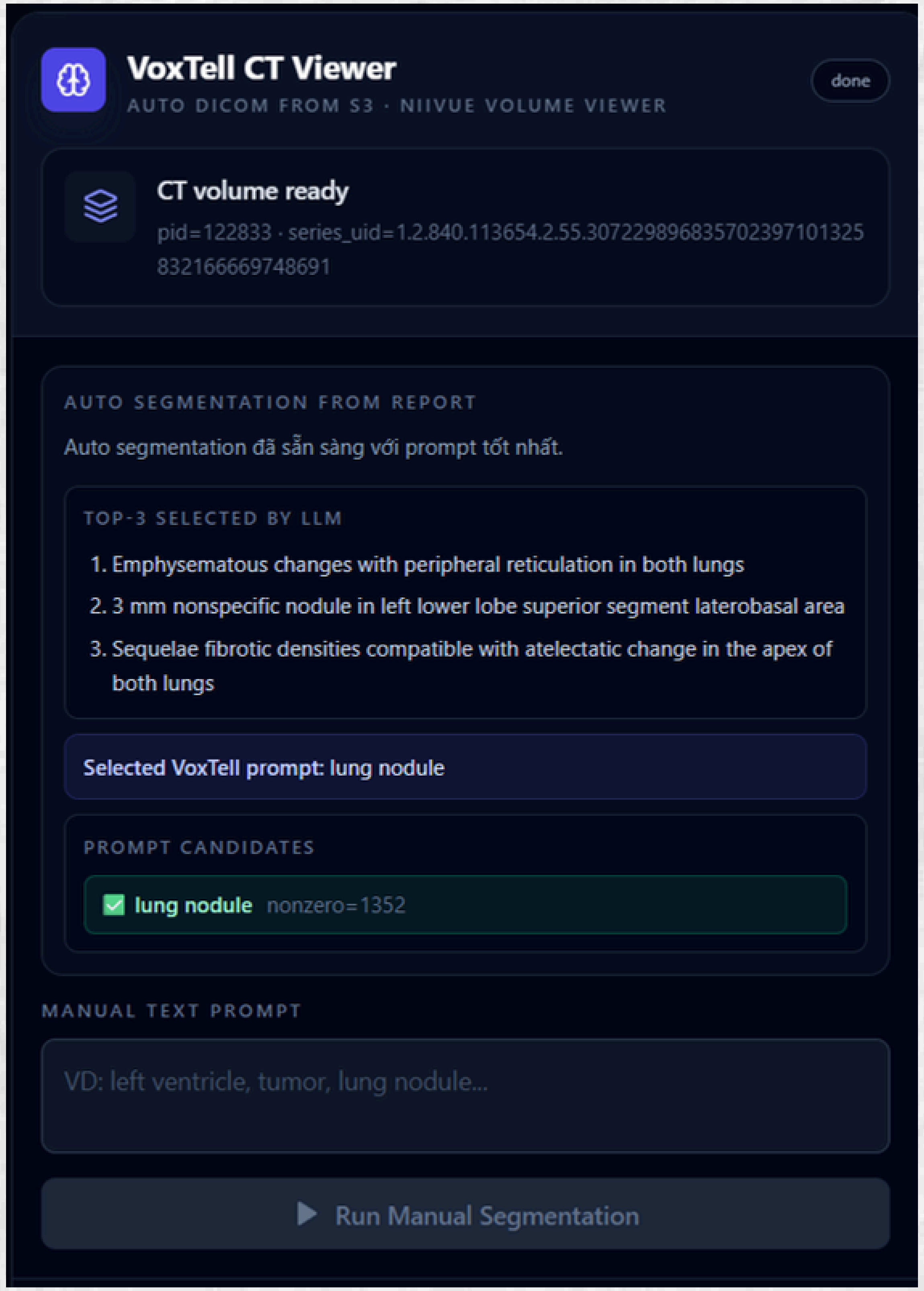

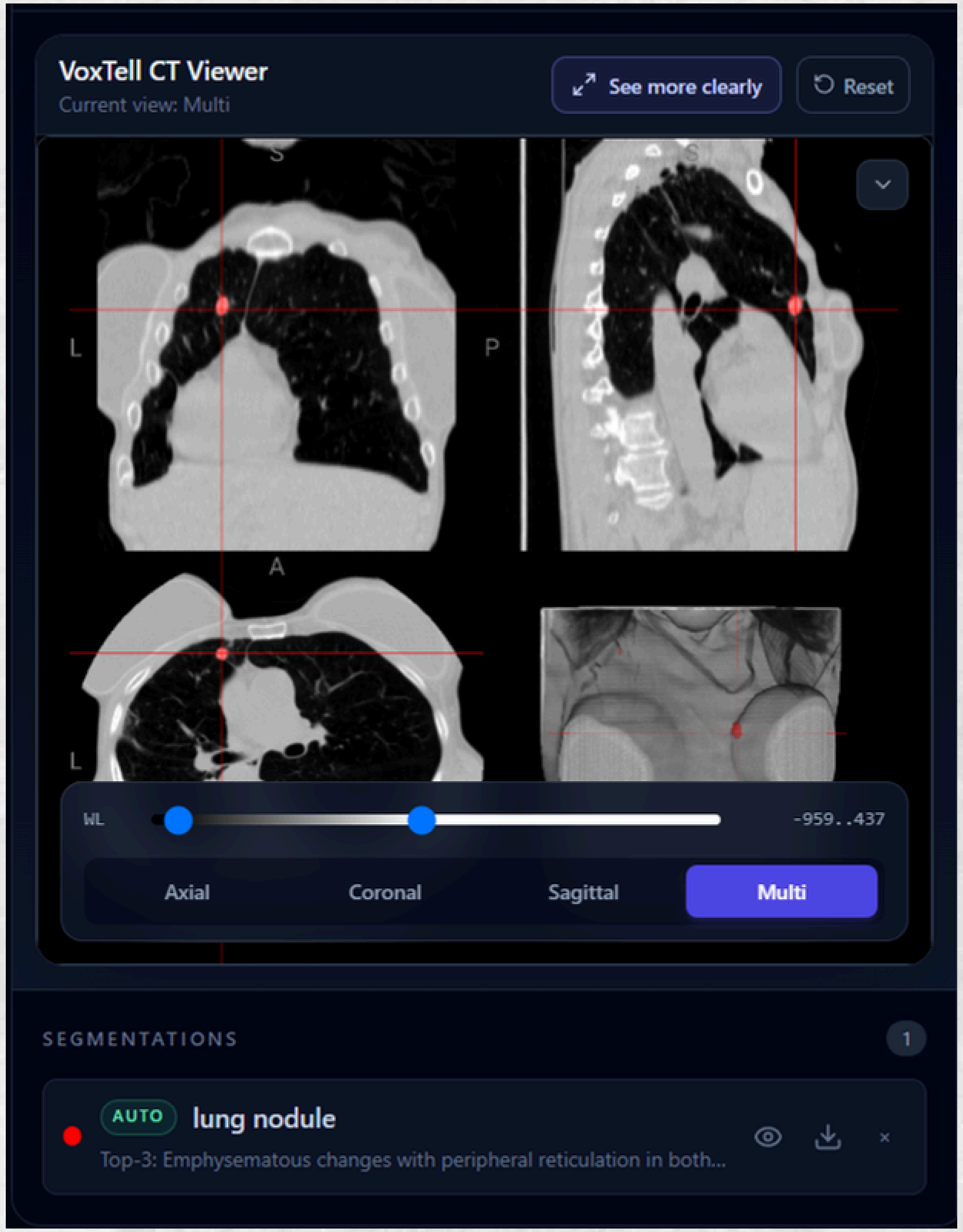

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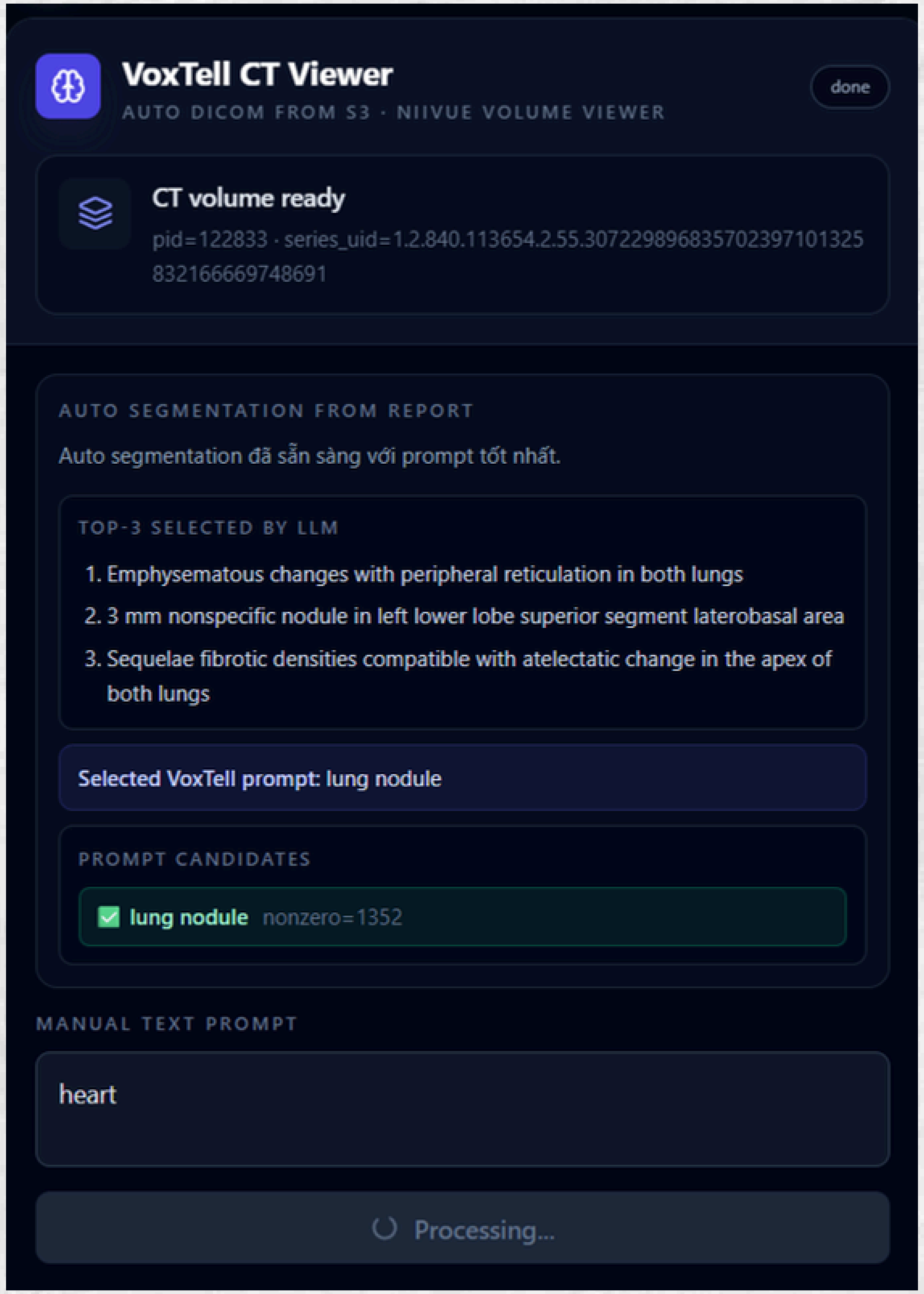

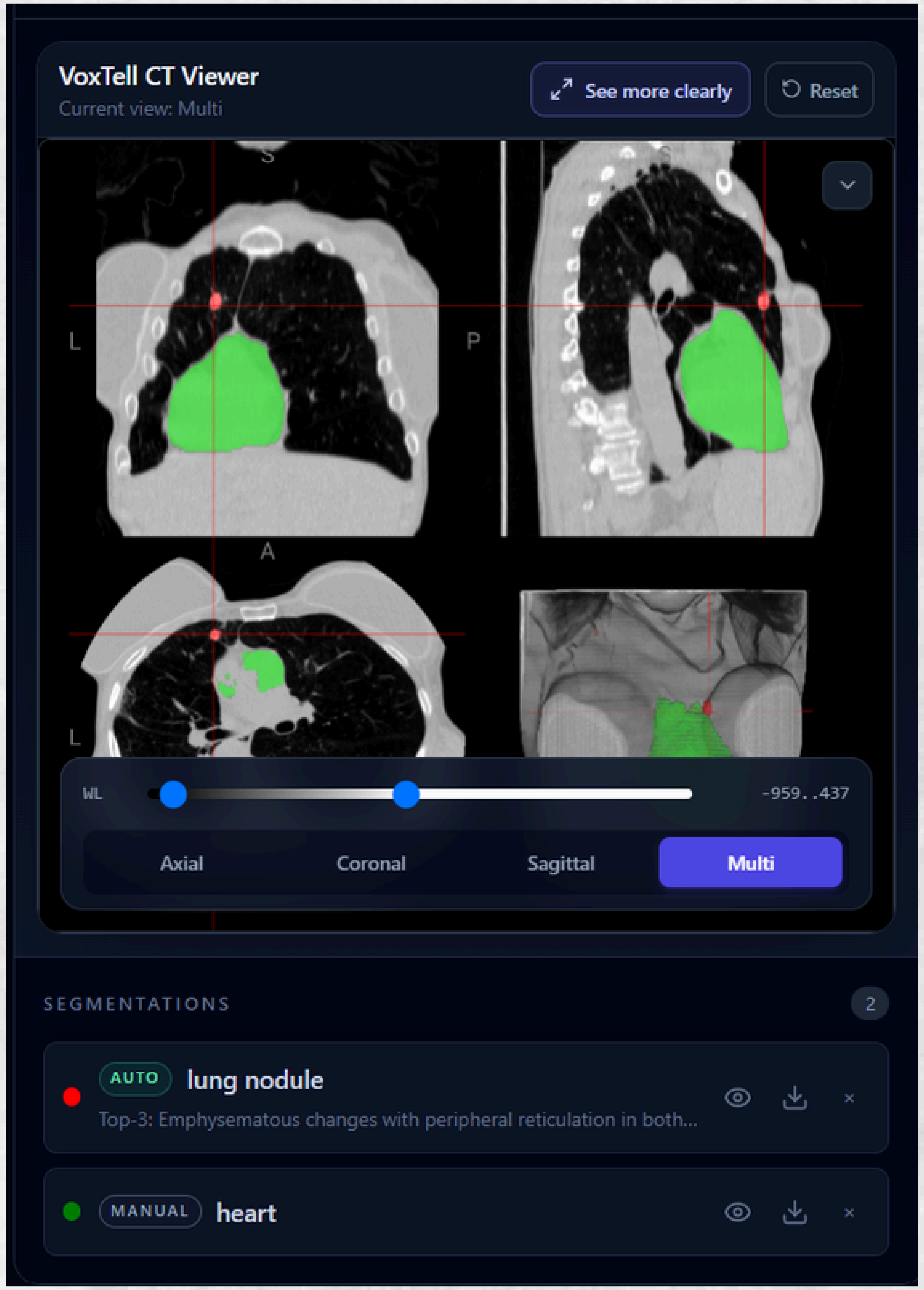

FUNCTION: SPEECH TO TEXT

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Purpose:

- Allow doctors to record their clinical comments or questions directly via a Microphone button in the browser.
- Automatically transcribe speech to specialized English medical text in real-time and insert the result into the input field for medical Chatbot queries.
- Help doctors minimize manual typing time during complex case consultations

FUNCTION: SPEECH TO TEXT

Model Specification:

- core model: OpenAI Whisper-Base model (73 million parameters) fine-tuned on a specialized medical terminology dataset focusing on Medicine and Surgery
- Inference Library: faster-whisper
- Resources Allocated: Runs entirely on CPU (cpu=2.0 - 2 vCPUs, memory=2048 - 2GB RAM).

RESULTS

RESULTS

Evaluation Metrics for Clinical Report Generation

Lexical Overlap Metrics (BLEU & ROUGE):

- **BLEU (1-4):** Measures n-gram precision. It checks how many words/phrases in the generated report exactly match the ground truth.
- **ROUGE (1, L):** Measures recall. It ensures the model doesn't miss key clinical information from the reference.

Semantic Similarity (BERT-Score):

- **The "Gold Standard":** Unlike BLEU/ROUGE, BERT-Score uses contextual embeddings to measure meaning.
- **Why it matters:** In medicine, "Heart is enlarged" and "Cardiomegaly is present" mean the same thing. BERT-Score recognizes this similarity even if the words are different.

Classification Accuracy:

- Used for specific diagnostic tasks (Oncology/Cardiology/Radiology) to measure the percentage of correct clinical predictions.

RESULTS

Findings and Impressions Agent

Model	Task	BLEU	BLEU-1	BLEU-4	ROUGE-1	ROUGE-2	ROUGE-L	BERT-SCORE
Finetuned	Findings	**0.0652**	**0.1121**	**0.0447**	**0.3815**	**0.2568**	**0.3015**	**0.9398**
	Impressions	**0.0307**	**0.2938**	**0.0397**	**0.1702**	**0.1073**	**0.1503**	0.9147
Base	Findings	0.0033	0.0337	0.0003	0.1595	0.0260	0.1037	0.9126
	Impressions	0.0000	0.2514	0.0000	0.0833	0.0159	0.0740	**0.9151**

- **Vocabulary & Structure (BLEU/ROUGE):** The Base model struggles heavily with medical reporting standards (BLEU-4 is **nearly 0.0000**). Fine-tuning successfully forces the model to adopt the correct terminology and narrative flow, multiplying ROUGE-1 and ROUGE-L scores significantly.
- **Clinical Meaning (BERT-Score):** Interestingly, the Base model achieves a high BERT-Score (~0.915). This indicates the base LLM correctly identifies the underlying pathologies, but simply lacks the domain-specific vocabulary to express them.
- **The Fine-Tuning Impact:** LoRA fine-tuning effectively acts as a "formatting and vocabulary bridge," teaching the AI how to write like a real radiologist without losing its core semantic intelligence.

RESULTS

Oncology Agent

Task / F1-Score	BASE (No Lora)	NO EXPLAIN (Finetuned)	WITH EXPLAIN (Finetuned + CoT)
Lung cancer risk	**0.52**	0.5	0.46

Slight Regression 📉

- Base accuracy experienced a minor drop (from **0.52** to **0.46** using CoT).
- **Conclusion**: Suggests potential localized overfitting, imbalanced training data, or noise introduced by the Chain-of-Thought prompting in this specific domain.

Cardinalogy Agent

Task / F1-Score	BASE (No Lora)	NO EXPLAIN (Finetuned)	WITH EXPLAIN (Finetuned + CoT)
CVD diagnosis	0.21	0.5	**0.55**
CVD mortality	0.1	**0.94**	0.78

Massive Improvement 📈

- Successfully overcame baseline limitations.
- **Accuracy** for CVD mortality surged dramatically (from **0.10** to **0.94**).
- **Conclusion**: The model effectively learned complex cardiovascular clinical patterns.

RESULTS

Radiology Agent

Task / F1-Score	BASE (No Lora)	NO EXPLAIN (Finetuned)	WITH EXPLAIN (Finetuned + CoT)
Chest abn 54	0.52	**1**	**1**
Chest abn 55	0.58	0.21	**1**
Chest abn 56	0.59	0.86	**0.97**
Chest abn 57	0.61	**1**	**1**
Chest abn 58	0.58	**0.97**	**0.97**
Chest abn 59	0.47	**0.59**	0.51
Chest abn 61	0.39	**0.7**	0.65
Nodule attenuation	0.06	**0.27**	0.18
Nodule location	0.06	**0.27**	0.18
Nodule margin	0.26	**0.73**	**0.73**
Nodule presence	0.29	**0.58**	0.51
Nodule size	0.04	**0.38**	**0.38**

RESULTS

Radiology Key Findings

- **Essential Specialization:** The Base model failed significantly in Nodule tasks (~0.04 - 0.06 accuracy). Fine-tuning was mandatory to achieve diagnostic capability.
- **The CoT Breakthrough:** Chain-of-Thought (With Explain) achieved Perfect Accuracy (1.0) across multiple Chest Abnormalities (54, 55, 57).
- **Reasoning Recovery:** Notably in Chest abn 55, the direct fine-tuned model dropped to 0.21, but CoT recovered it to 1.0, proving that explicit reasoning prevents logical collapses.
- **Pattern vs. Logic:** No-Explain performed better on visual-spatial tasks (Nodules), while CoT excelled in complex diagnostic logic.

The Power of Fine-Tuning

- **Fine-Tuning > Base Model:** Across all domains (Oncology, Cardiology, Radiology), fine-tuning transformed a non-functional base model into a specialized medical expert.
- **Clinical Viability:** While Base models struggle with zero-shot medical reasoning, LoRA fine-tuning provides the domain-specific knowledge necessary for real-world clinical use.
- **Final Verdict:** Despite minor variances in Oncology, the massive accuracy gains (up to 9x improvement) in specialized tasks prove that domain-specific fine-tuning is the only path to reliable AI diagnostics.

DISCUSSION

CLINICAL VALUE OF MULTI-AGENT DISCREPANCIES

How Agent Conflicts Empower Clinicians

1. Eliminating AI Confirmation Bias

- Single-model AI often forces doctors into "blind trust."
- In our system, discrepancies between the Base Agents and the Verification Agent serve as an automated clinical red flag, preventing physicians from overlooking critical omissions.

2. Active Human-in-the-Loop Triage

- When agents disagree, the system does not make an arbitrary choice. Instead, it flags the case as a High-Risk Mismatch.
- This mismatch instantly alerts the doctor to actively intervene and utilize Feature 2 (Interactive Segmentation) or Feature 3 (RAG Chat) to audit the exact medical pixels.

3. Multi-Perspective Diagnostic Safety

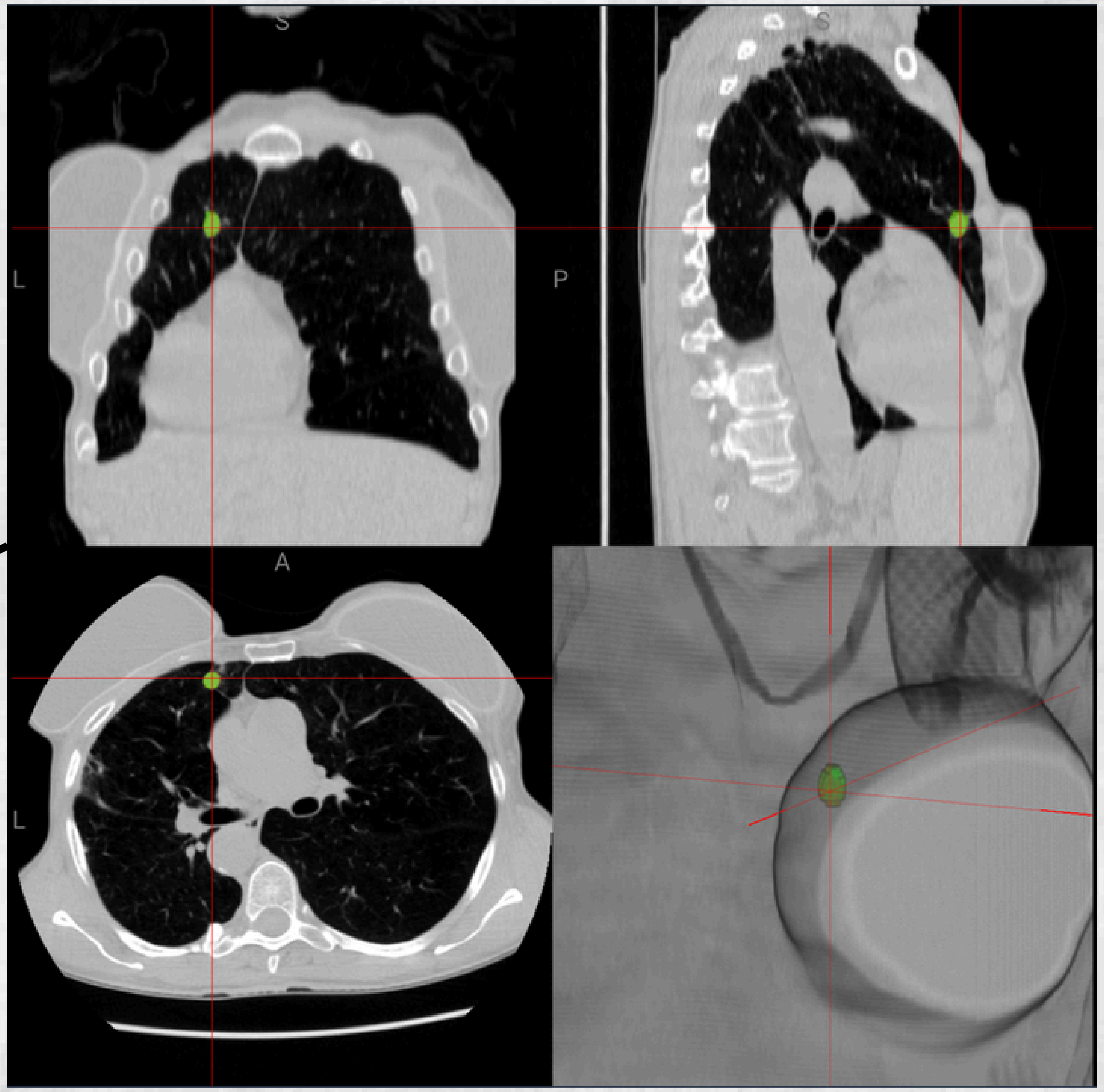
- By exposing the internal conflict (e.g., text vs. image, focal vs. diffuse), the framework provides doctors with a comprehensive, multi-angle view of the patient's condition, converting technical boundaries into clinical safety nets.

CLINICAL EVALUATION - CASE 1: LUNG NODULE

COMPONENT	RADIOLORY AGENT	VERIFICATION AGENT	GROUND TRUTH
Lung Nodule Detail	• **Location**: Right Lower Lobe • **Size**: 4~6mm • **Margin**: Smooth	• **Correction:** Relocated to the Left Lung • **Refinement:** Solitary nodule is < 6mm (~3mm)	• **Location:** Left Upper Lobe • **Size:** 6~8mm • **Margin:** Poorly defined

The Conflict: Base agents flag a smooth nodule in the Right Lower Lobe, while the Verification agent identifies it in the Left Lung. Meanwhile, Ground Truth marks a dangerous Poorly-defined nodule in the Left Upper Lobe.

"Segment the poorly defined solid nodule in the left upper lobe."



CONCLUSION

CONCLUSION

- This project successfully develops a web-based medical AI system for supporting chest CT analysis.
- The system integrates DICOM preprocessing, clinical data processing, LoRA fine-tuned agents, report generation, verification, and lesion segmentation.
- Experimental results show that LoRA fine-tuning significantly improves model performance, especially for the Radiology Agent, Cardiology Agent, and Findings and Impressions Agent.
- Fine-tuning not only improves prediction performance but also helps the model generate reports with more appropriate medical terminology and structure.

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TECHSTACK

Modal — CLOUD & SERVERLESS — Microsoft Azure

WEB APPLICATION

WEB CLIENT

HTML CSS JS

BACKEND

FastAPI python™

AI

vLLM

unsloth

LangChain

DATABASE

PostgreSQL

Amazon S3

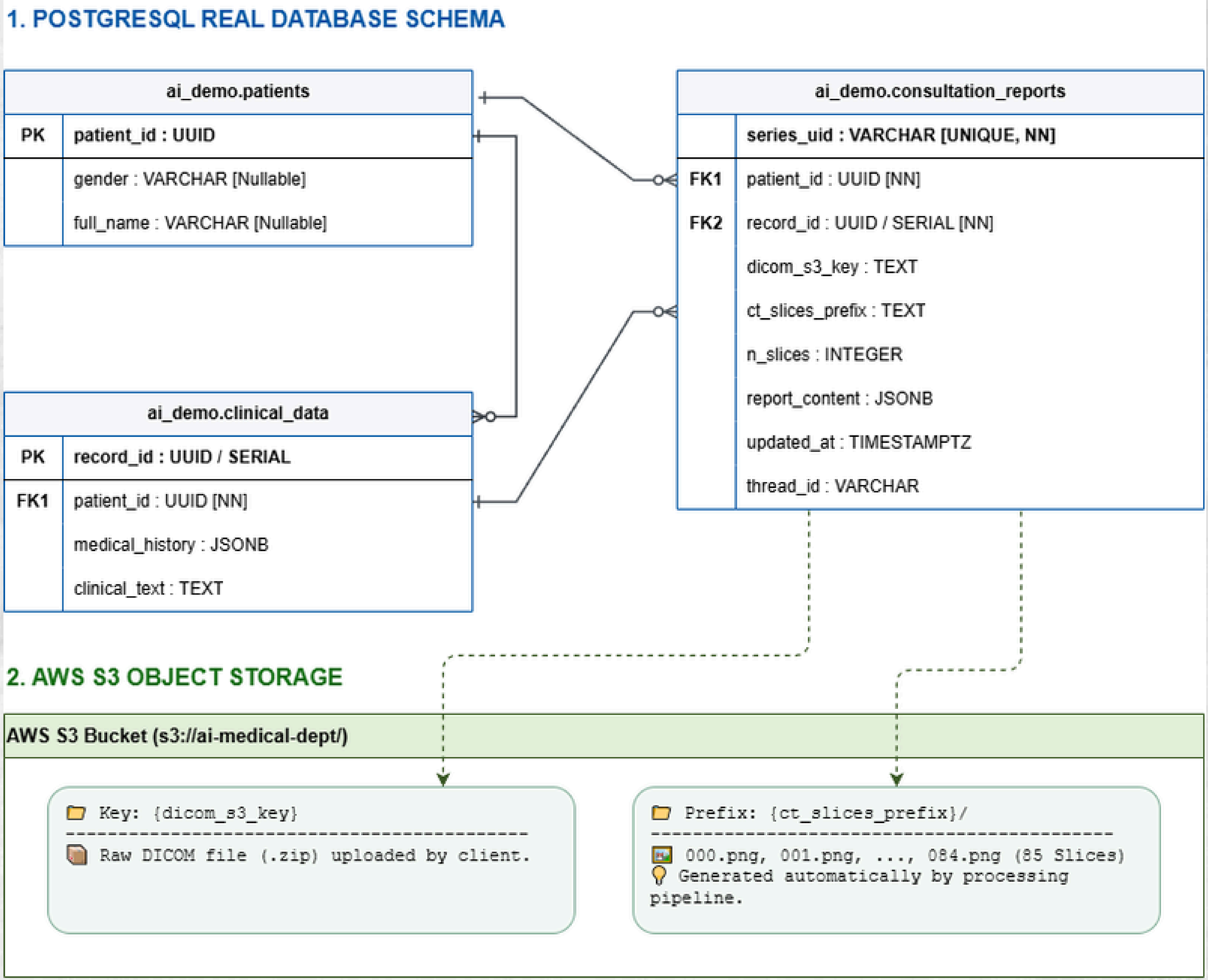
EXTERNAL SERVICE

deepseek

HUGGING FACE

GitHub

DATABASE DESIGN



THANK YOU

FOR LISTENING

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